

CIS 635: Data Mining and Knowledge Discovery

Team Project: Guidelines and Requirements

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Each student completes a data mining and knowledge discovery (KDD) project, working in a team of up to two students.

The project applies the concepts, tools, and techniques covered in this course to a real, multi-dataset problem — following the KDD process from data understanding through evaluation and communication of results.

Grade weight
30%
of total course grade

Team Formation

- Teams may consist of **one or two students**, with two-student teams **preferred**
- Students select their own teammates; the instructor can help match students who need a partner upon request
- Team composition must be **finalized and reported** to the instructor before the Project Specification Document is submitted in **Week 4**

Project Timeline and Deliverables

When	Deliverable / Milestone	Notes
Week 4	Project Specification Document due	Sample provided in the Appendix
By Wk 5	Development tool validated (if not using Colab)	
Wks 5–7	Progress Check-In #1	Brief meeting with instructor
Wks 9–11	Progress Check-In #2	Brief meeting with instructor
Wks 13–14	Check-In #3 and Final Presentation	
Week 14	Final Project Report due	

- **Google Colab** is the primary/default environment for all teams
(Python-based notebooks, free GPU access, no local setup required)
- Teams may propose an alternative (e.g., local Jupyter, RStudio, a specific cloud platform)
— submitted for instructor review and **validated no later than Week 5**
- Tool-change requests should briefly state the proposed tool and why it is needed

Default assumption

Until an alternative is approved in writing, teams should proceed as if Colab is required.

Dataset Requirements

Each project must use **at least two genuinely inter-related datasets**: sharing a meaningful connection (key, entity, geography, time period, or subject) that lets them be **joined or combined** into a single analysis.

Datasets analyzed merely side by side do not meet this requirement. Any exception must be approved by the instructor.

What Counts as “Inter-related”

The datasets must be linkable through one or more of the following — and the project must **actually use** that link (join, merge, or shared grouping variable):

- A shared identifier or key (e.g., patient ID, sample ID, customer ID, product ID)
- A shared geographic dimension (e.g., county, state, ZIP code, region)
- A shared temporal dimension (e.g., matching date ranges or reporting periods)
- A logical/contextual relationship — one dataset provides features, labels, or context for records in the other

Acceptable (genuinely related)

- Gene-expression data + clinical/phenotype metadata (shared sample ID)
- Patient/clinical data + county-level public-health data (shared geography)
- Clinical trials registry + adverse-event data (shared trial ID)
- Customer transactions + customer demographics (shared customer ID)

Not acceptable

Two unrelated public datasets from different domains, each analyzed on its own with no shared key, join, or combined analysis.

Dataset Ethics and Privacy

- All datasets must be **publicly available**, properly licensed, or otherwise approved in advance for academic use
- Must not contain PII or PHI beyond what is already de-identified in a public, approved repository
- **No IRB approval** is required for existing, publicly available, de-identified datasets on this course project
- A restricted or identifiable dataset (data-use agreement, PHI) requires consulting the instructor first — generally not appropriate for this timeline
- Every dataset used must be **cited** in the specification document and final report, including source and license/terms of use

Progress Check-Ins

Each team meets briefly with the instructor three times during the semester — short, informal check-ins rather than written reports.

Check-in	Window	Focus
#1	Wks 5–7	Confirmed datasets and relationship, initial exploration/cleaning, refined scope
#2	Wks 9–11	Feature engineering/preprocessing complete, initial modeling or analysis results
#3	Wks 13–14	Final results, remaining work, presentation readiness (paired with the final presentation)

Teams sign up for a meeting slot within each window; come prepared with a one-paragraph status update.

Suggested structure, following the KDD process:

1. Introduction and problem motivation
2. Data description
3. Data preprocessing and integration
4. Methodology
5. Results
6. Discussion and limitations
7. Conclusion and future work
8. References and dataset citations
9. Appendix: link to the Colab notebook

- Delivered during the **Week 13–14** window, alongside Check-In #3
- **Suggested length:** 10–15 minutes per team, including questions
- Should summarize: the problem, datasets and their relationship, methods, key results, and lessons learned

Grading Breakdown

This project counts for **30% of the course grade.**

Component	Weight	
Project Specification Document (Wk 4)	3%	of 30%
Progress Check-Ins (3, combined)	7%	of 30%
Final Project Report	10%	of 30%
Final Presentation	10%	of 30%
Total	30%	of course grade

Appendix: Sample Specification Document

A fully worked example (team of two, 6–8 weeks):

Predicting Breast Cancer Molecular Subtype from Linked Gene-Expression and Clinical Metadata

- **Datasets:** a GEO gene-expression matrix + a clinical/phenotype metadata table for the same samples — linked by shared GSM sample accession
- **Methods:** preprocessing & feature selection/PCA; supervised classification (logistic regression, random forest); unsupervised clustering (k-means/hierarchical); evaluation (accuracy/F1/ROC-AUC; silhouette & cross-tabulation)
- **Tools:** Google Colab — pandas, NumPy, scikit-learn, seaborn/matplotlib
- **8-week work plan:** acquire & join → EDA/cleaning → feature selection/PCA → baseline model → clustering → tuning → draft report → finalize (buffer)

Provided as a model for structure and scope — teams replace it with their own topic, not reuse it.

Key Takeaways

Remember:

- Team finalized before Week 4
- Two **genuinely inter-related** datasets — the join is the point
- Colab is the default; alternatives need Week 5 approval
- Cite every dataset's source and license
- Three check-ins: Wks 5–7, 9–11, 13–14
- Final report + presentation, Week 14

Questions?

Full details: Team Project Guidelines document (supplements the syllabus)